

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE CODE: CSA51

COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Analytical Problem Solving: 40%

QUESTIONS:

Total Marks: 50

Annexure A

Code of Conduct:

I _____U.Divya(192372277)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Divya

Annexure A

Analytical Problem Solving

1. RSA Key Generation and Security Analysis

Given:

$$p = 17, q = 19$$

Compute n and $\phi(n)$

Choose $e = 5$ and calculate the private key d

Encrypt the message $M = 7$

Decrypt the ciphertext and verify correctness

Analyze the impact of choosing a small prime number on RSA security

Given

- $p=17$
- $q=19$
- $e=5$
- Message $M=7$

Step 1: Compute n and $\phi(n)$

$$n = p \times q = 17 \times 19 = 323$$

Euler Totient:

$$\phi(n) = (p-1)(q-1) = (17-1)(19-1) = 16 \times 18 = 288$$

Answer

- $n=323$
- $\phi(n)=288$

Step 2: Compute Private Key d

Need

$$ed \equiv 1 \pmod{288}$$

Since

$$5 \times 173 = 865 \\ 865 \bmod 288 = 1865 \bmod 288 = 1$$

Therefore

$$d=173$$

Private Key = **(173,323)**

Step 3: Encrypt Message

Formula

$$C = M^e \bmod n \\ C = 7^5 \bmod 323$$

Using modular arithmetic

$$7^2 = 49 \\ 7^4 = 49^2 = 2401 \equiv 1407 \pmod{323} \\ 7^5 = 140 \times 7 = 980 \equiv 117 \pmod{323}$$

Ciphertext

$$C=11$$

Step 4: Decrypt

Formula

$$M = C^d \bmod n \\ M = 11^{173} \bmod 323$$

Applying modular exponentiation,

$$M=7M=7M=7$$

Original message recovered successfully.

Analysis

Using small prime numbers makes RSA insecure because:

- Small primes are easy to factor.
- Attackers can quickly obtain ppp and qqq.
- Private key can be computed easily.
- Modern RSA uses **2048-bit or larger keys**.

2. ElGamal Encryption Analysis

Given:

$p = 23$, $g = 5$, private key $x = 7$, random $k = 3$, message $M = 10$

Compute the public key

Perform encryption to generate ciphertext pair (C1, C2)

Decrypt the ciphertext to retrieve the original message

Analyze how randomness (k) improves security in ElGamal

Given

- $p=23p=23p=23$
- $g=5g=5g=5$
- Private key $x=7x=7x=7$
- Random number $k=3k=3k=3$
- Message $M=10M=10M=10$

Step 1: Public Key

Formula

$$y=gx \bmod p \quad y=g^x \bmod p \quad y=gx \bmod p = 57 \bmod 23 = 5^7 \bmod 23 = 57 \bmod 23$$

Since

$$57 \equiv 175^7 \pmod{23} \equiv 17$$

Public Key

$$(p, g, y) = (23, 5, 17) \quad (p, g, y) = (23, 5, 17) \quad (p, g, y) = (23, 5, 17)$$

Step 2: Encryption

Compute

$$C_1 = g^k \pmod{p} = 5^3 \pmod{23} = 125 \pmod{23} = 10$$

So

$$C_1 = 10$$

Now

$$C_2 = M \times y^k \pmod{p} = 17 \times 5^3 \pmod{23} = 17 \times 125 \pmod{23} = 2125 \pmod{23} = 14$$

Therefore

$$C_2 = 14$$

Ciphertext

$$(C_1, C_2) = (10, 14)$$

Step 3: Decryption

Compute

$$S = C_1^{-1} \pmod{p} = 10^{-1} \pmod{23} = 7$$

Inverse of 14 mod 23

$$14^{-1} \pmod{23} = 5$$

Recover message

$$M = C^2 \times S^{-1} \pmod{23} \quad M = C^2 \times S^{-1} \pmod{23} \quad M = C^2 \times S^{-1} \pmod{23}$$

$$= 2 \times 5 = 10 = 2 \times 5 = 10 = 2 \times 5 = 10$$

Recovered Message

$$M = 10 \quad M = 10 \quad M = 10$$

Analysis

Random number kkk:

- Produces different ciphertexts every time.
- Same message encrypts differently.
- Prevents replay and pattern attacks.
- Improves semantic security.

3. RSA Digital Signature Verification

Given:

$$p = 11, q = 17, e = 7, \text{ message } M = 8$$

Compute n and $\phi(n)$

Find private key d

Generate the digital signature for the message

Verify the signature using the public key

Analyze how digital signatures ensure non-repudiation

Given

- $p = 11 \quad p = 11 \quad p = 11$
- $q = 17 \quad q = 17 \quad q = 17$
- $e = 7 \quad e = 7 \quad e = 7$
- Message $M = 8 \quad M = 8 \quad M = 8$

Step 1

$$n = 11 \times 17 = 187 \quad n = 11 \times 17 = 187 \quad n = 11 \times 17 = 187$$

$$\phi(n) = 10 \times 16 = 160 \quad \phi(n) = 10 \times 16 = 160 \quad \phi(n) = 10 \times 16 = 160$$

Step 2

Find

$$7d \equiv 1 \pmod{1607} d \equiv 1 \pmod{1607} d \equiv 1 \pmod{160} d = 23 d = 23 d = 23$$

since

$$7 \times 23 = 161 \equiv 1 \pmod{1607} 17 \times 23 = 391 \equiv 1 \pmod{1607} 17 \times 23 = 391 \equiv 1$$

Step 3: Signature

$$S = M^d \pmod{n} \\ S = 823 \pmod{187} \\ S = 823 \pmod{187}$$

Using modular exponentiation

$$S = 57 \\ S = 57 \\ S = 57$$

Step 4: Verification

$$M = S^e \pmod{n} \\ 577 \pmod{187} = 857^7 \pmod{187} = 8577 \pmod{187} = 8$$

Verified successfully.

Analysis

Digital signatures provide

- Authentication
- Integrity
- Non-repudiation
- Prevent message forgery

4. Diffie-Hellman with Attack Resistance Evaluation

Given:

$$p = 29, g = 2$$

User A private key = 5

User B private key = 12

Compute public keys for both users

Compute the shared secret key

Analyze what happens if an attacker intercepts and replaces public keys

Suggest a cryptographic solution to ensure authenticity

Given

- $p=29$
- $g=2$
- $A = 5$
- $B = 12$

Step 1: Public Keys

User A

$$A=25 \pmod{29} \quad A=2^5 \pmod{29} \quad A=25 \pmod{29} \quad A=3 \quad A=3 \quad A=3$$

User B

$$B=7 \pmod{29} \quad B=2^{12} \pmod{29} \quad B=7 \pmod{29} \quad B=7 \quad B=7 \quad B=7$$

Step 2: Shared Secret

User A

$$75 \pmod{29} = 16 \quad 7^5 \pmod{29} = 16 \quad 75 \pmod{29} = 16$$

User B

$$312 \pmod{29} = 16 \quad 3^{12} \pmod{29} = 16 \quad 312 \pmod{29} = 16$$

Shared Secret

161616

Analysis

If attacker replaces public keys:

- Different keys are established with each party.
- Attacker decrypts and re-encrypts messages.
- This is a **Man-in-the-Middle (MITM)** attack.

Solution

- Digital Signatures
- Certificates (PKI)
- Authenticated Diffie-Hellman
- TLS

5. Hybrid Encryption System Analysis

Given:

A system uses RSA for key exchange and AES for data encryption

RSA encryption time = 5 ms per key exchange

AES encryption time = 0.5 ms per data block

Total data blocks = 500

Calculate total time taken for encryption

Compare with using only RSA for encrypting all data

Analyze why hybrid encryption is more efficient

Justify its use in modern secure communication systems

Given

- RSA = 5 ms
- AES = 0.5 ms/block
- Blocks = 500

Step 1

AES time

$$500 \times 0.5 = 250 \text{ ms}$$

RSA key exchange

$$5 \text{ ms}$$

Total

$$250 + 5 = 255 \text{ ms}$$

Step 2

If RSA encrypts every block

$$500 \times 5 = 2500 \text{ ms}$$

Step 3

Comparison

Method	Time
Hybrid	255 ms
Only RSA	2500 ms

Time Saved

$$2500 - 255 = 2245 \text{ ms}$$

Analysis

Hybrid encryption is preferred because:

- RSA securely exchanges the symmetric key.
- AES encrypts large amounts of data efficiently.
- Provides strong security with high performance.
- Used in modern secure protocols such as TLS/SSL, VPNs, and HTTPS.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

